Milestone 4: Progress Update - PathMate.ai

Project Live Link: https://career-helper-dun.vercel.app/

GitHub Repository: https://github.com/HEEESANGKIM/career-helper

Date: May 31, 2026

Team Members: Dohyeon Lee, Heesang Kim

1. Individual Progress Update

Dohyeon Lee

- Scheduled Tasks (from Software Design):
    - **[M1]** Refine UML architectural diagrams, focusing on final updates for Class and Sequence Diagrams
    - **[M2]** Create global reusable common UI components
    - **[M2-M3]** Develop user interfaces for the Career Cockpit Dashboard and Profile Management pages
    - **[M3]** Create poster presentation materials
    - **[M3]** Build the Interactive AI Workspace, including the Hiring Simulator and AI Cover Letter Editor
    - **[M4]** Participate in final system integration and deployment to the Vercel production environment

- Completed:
    - **[M1]** Optimized and finalized UML Class and Sequence Diagrams to precisely model system behavior and data flows — *Completed on May 5, 2026*
    - **[M2]** Built and verified global reusable common UI components, including button and input form libraries — *Completed on May 12, 2026*
    - **[M2-M3]** Implemented core user interfaces (Homepage, Profile Management, and Dashboard) using Next.js and Tailwind CSS, and successfully merged them into the main branch — *Completed on May 24, 2026*
    - **[M3]** Authored and finalized poster session materials for offline presentation and evaluation — *Completed on May 17, 2026 (Early Submission)*
    - **[M3]** Constructed interactive UIs for the Hiring Simulator and AI Cover Letter Workspace, and integrated high-fidelity dummy data — *Completed on May 26, 2026*

- In-Progress:
    - [M4 Check-in] Conducting cross-testing for overall system functionalities and leading the resolution of UI/UX layout bugs tracked via GitHub Issues.

Heesang Kim

- Scheduled Tasks (from Software Design):
    - **[M1]** Finalize Software Design Document (SDD) structure and organize core text sections
    - **[M1]** Establish detailed backend API specifications and endpoint mappings via Google Sheets
    - **[M2]** Construct database API endpoints for Profile CRUD operations and Dashboard statistics
    - **[M3]** Design AI backend logic: Integrate OpenAI API and enhance personalization matching algorithms
    - **[M3]** Create poster presentation materials
    - **[M4]** Optimize system logic: Advance AI prompt engineering and implement robust exception handling
    - **[M4]** Lead the final system integration and project deployment to the Vercel production environment

- Completed:
    - **[M1]** Finalized the Software Design Document (SDD) structure, drafting the core introduction and system architecture sections — *Completed on May 3, 2026*
    - **[M1]** Completed detailed RESTful API design specifications, including OpenAI and Supabase endpoint mappings on Google Sheets — *Completed on May 6, 2026*
    - **[M2]** Designed the comprehensive database schema and established foundational CRUD API endpoints leveraging Supabase — *Completed on May 10, 2026*
    - **[M3]** Integrated core AI engines using Next.js Server Actions and the OpenAI SDK to generate Fit Scores, real-time interview questions, and automated cover letter drafts — *Completed on May 22, 2026*
    - **[M3]** Authored and finalized poster session materials, capturing system architectures and detailed data flow diagrams — *Completed on May 17, 2026 (Early Submission)*

        - **[M4]** Optimized backend system logic, advanced AI prompt instructions, and established matching logic exception handling — *Completed on May 28, 2026*
        - **[M4]** Led final multi-module system integration and executed the live Beta release of PathMate.ai on Vercel — *Completed on May 31, 2026*

- In-Progress:
    - [M4 Check-in] Performing cross-testing on frontend components, managing the continuous re-deployment pipeline to Vercel incorporating bug-fix updates from Dohyeon, and optimizing database/cloud infrastructure.

2. Group Progress Update

■ Schedule Comparison

Overall, the project is progressing smoothly and systematically in alignment with our baseline milestones. Following the early finalization of our presentation materials on May 17th and the successful live showcase at the Poster Presentation Session on May 27th (Milestone 3), the team has seamlessly transitioned into the deployment phase, finalizing the multi-module system integration and successfully launching the official website Beta version via Vercel by May 31st (Milestone 4). With all core use cases fully operational, verified, and merged into the main branch exactly on schedule, we have dynamically shifted our focus into the comprehensive cross-validation, performance optimization, and bug-tracking workflows of the Final Release phase (Milestone 5) to prepare for the upcoming final evaluation deliverables.

■ Current Status & Self-Assigned Grade

The project currently stands at an **A-** grade level, safely satisfying the core Milestone 4 evaluation guidelines by delivering a fully functional, live Beta platform where all primary use cases are completely integrated and accessible via a public production link. Although we experienced a minor, non-blocking schedule adjustment while refining asynchronous exception-catching and response latency within the OpenAI API backend, our overall implementation is highly stable. Having successfully showcased our system architecture early at the live Poster Presentation Session and seamlessly deployed the operational end-to-end framework on Vercel, this grade reflects the team's strong engineering execution, thorough cross-validation, and high readiness for the upcoming final release phase.

■ Issue Analysis & Adjustment Plan

- Challenges Identified: During the mutual cross-validation of our integrated modules, we identified several minor edge-case anomalies, particularly regarding exceptional user inputs such as blank field submissions and intermittent OpenAI API gateway timeouts. These constraints occasionally induced temporary client-side interface hanging or partial asynchronous data omission. While these bugs are non-blocking and do not compromise the core application architecture, systematically handling these client-side exceptions and optimizing rendering fail-safes emerged as our primary operational challenge moving into the stabilization phase.

- Adjustment Plan: To systematically resolve these issues under our accelerated semester timeline, the team has formally established a dedicated Quality Assurance (QA) track within our updated Milestone 5 (Final Release) schedule, ensuring that every bug is registered in real-time via GitHub Issues and rigorously verified through continuous mutual cross-testing. Under this structured framework, we have optimized our division of labor: Dohyeon Lee is fully dedicated to refactoring client-side exception code, adding loading animations, and fixing frontend layout bugs to maximize UI/UX stability, while Heesang Kim manages live codebase merges, executes production hot-fixes on Vercel, and optimizes database indexing. This proactive workflow expansion into Milestone 5 ensures that the final integrated system undergoes comprehensive end-to-end regression testing, guaranteeing maximum runtime performance and data integrity before the final presentation and report submission.

3. Schedule & Remainder

<u>Milestone 1: Design Review (Completed: Early May)</u>
Goal: Complete Software Design Document and determine system architecture.

Tasks:

- Software Design Document Finalization: Complete Document Structure and Text Organization (Heesang Kim) — *Completed on May 3, 2026*
- UML Diagram Precision: Class and Sequence Diagram Final Update (Dohyeon Lee) — *Completed on May 5, 2026*
- API Design Completed: Detailed API Specification with Google Sheet (Heesang Kim) — *Completed on May 6, 2026*

<u>Milestone 2: Alpha Release (Completed: Mid-May)</u>
Goal: Build basic infrastructure and complete User Interface framework.

Tasks:
- Data API endpoints: Profile data CRUD and dashboard statistics deployment API (Heesang Kim) — *Completed on May 11, 2026*
- Common UI components: Create a button, input form library for reuse across the globe (Dohyeon Lee) — *Completed on May 13, 2026*
- Core UI Layout: Career Cockpit Dashboard and Profile Management Page UI Development (Dohyeon Lee) — *Completed on May 15, 2026*

Milestone 3: Poster Presentation (Completed: Late May)

Goal: Visualize system architecture, complete the poster showcase, and gather initial peer evaluation.

Tasks:

- Poster presentation material production: Poster completion visualizing system architecture and core flows (Together) — *Submitted on May 17, 2026*
- AI Backend Logic: OpenAI API integration matching algorithm and interview question generation engine enhancement (Heesang Kim) — *Completed on May 22, 2026*
- Interactive AI workspace: Interview simulation interface and cover letter editing workspace check (Dohyeon Lee) — *Completed on May 24, 2026*
- Poster Presentation Showcase: Conducted the live poster presentation session and collected initial peer evaluation (Together) — *Presented on May 27, 2026*

Milestone 4: Beta Release & Progress Check-In (Completed: Early June)

Goal: Logic optimization, final multi-module system integration, and official website beta deployment.

Tasks:

- Logic Optimization: AI Prompt Engineering Advancement and Matching Logic Exception Handling (Heesang Kim) — *Completed on May 28, 2026*
- Integration & Vercel Deployment: Final system integration, deployment, and official Milestone 4 Beta Release via Vercel (Together) — *Completed on May 31, 2026*
- Milestone 4 Progress Update Documentation**:** Compile individual and group progress reports for professor check-in (Together) — *Completed on June 1, 2026*

Milestone 5: Final Release (In-Progress / Remainder Schedule)

Goal: Reflect user feedback, eliminate outstanding bugs, optimize performance, and deliver final deliverables.

Tasks:

- [In-Progress / Adjusted QA Workflow] Comprehensive Bug Tracking & Final Check: Execute mutual cross-validation based on Beta feedback. Dohyeon is tracking and fixing frontend layout bugs via GitHub Issues while Heesang handles continuous production re-deployments and data integrity checks (Together) — *Scheduled: May 29 – June 4, 2026*
- [In-Progress] Optimizing and Securing Performance: Improved API response speed, enhanced database indexing, and data security reinforcement (Heesang Kim) — *Scheduled: May 30 – June 4, 2026*
- [In-Progress] UI/UX Polishing: Add smooth micro-animations, improve asynchronous loading statuses, and conduct a final mobile responsive layout check (Dohyeon Lee) — *Scheduled: May 30 – June 4, 2026*
- Final Presentation: Supplement final functionality based on collected feedback and deliver the final live project presentation to the instructor (Together) — *Target Date: June 5, 2026*
- Final Report: Documentation of project performance and future development directions (Together) — *Target Date: June 10, 2026*